# Stacking Galaxies Reveals a Universal Baryon Fountain

*Averaging many simulated galaxies turns messy, individual gas flows into one clean, coherent cycle.*

**arXiv:2609.21374 — Stacked baryon cycles in cosmological hydrodynamic simulations**

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## Background

Galaxies grow by pulling gas in and pushing it back out, forming fountain-like cycles that shape their evolution. Individual systems are too diverse and dynamic to reveal the underlying pattern directly.

**Observational hints of baryon cycles exist, but no clean, unified picture has emerged from messy individual galaxies.**

## What this paper is selling

The authors stack gas fields across many simulated galaxies, suppressing noise to expose strikingly regular inflow and outflow patterns. The resulting fountain model matches observations and compresses the cycle into a few measurable parameters.

## Key results to remember

- Stacking reveals smooth, coherent gas velocity fields across entire galaxy ecosystems.
- Outflow cones and fountain geometry persist across stellar masses and cosmic time.
- A simple parameterized fountain model reproduces the stacked velocity field.

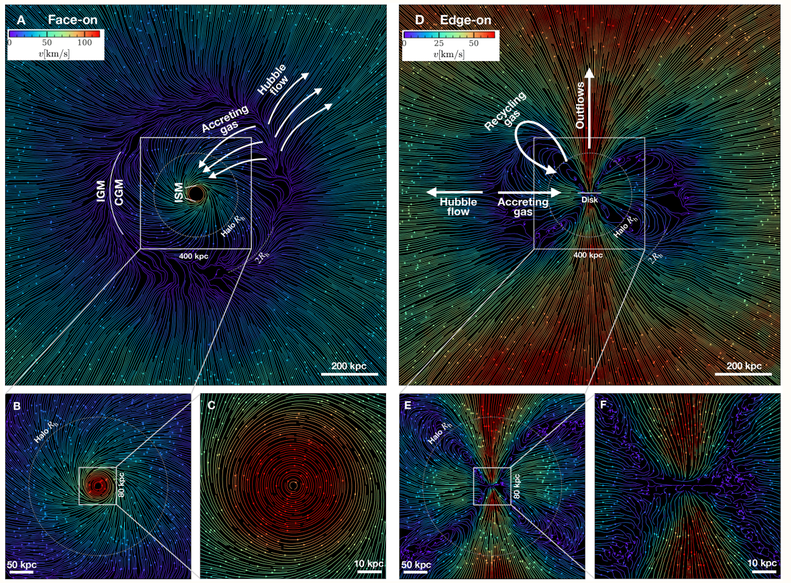



**Figure 1 |** Stacked streamlines show a smooth, coherent gas velocity field, with dashed circles marking the median virial radius.

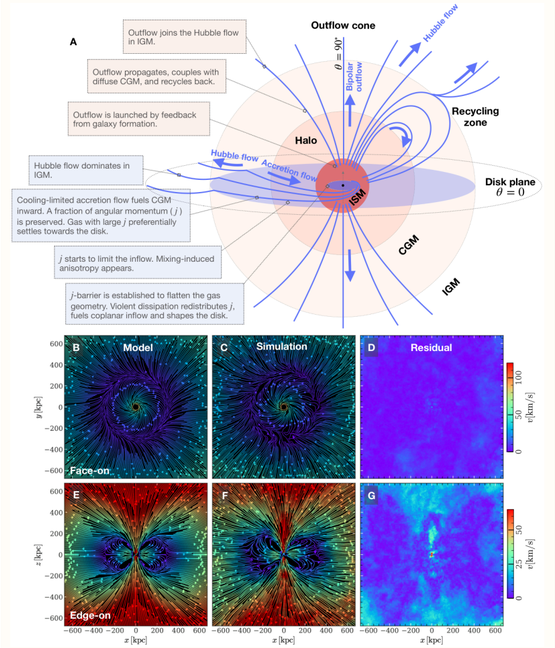



**Figure 4 |** The parameterized fountain model matches the simulated velocity field, with residuals small enough to quantify the cycle.